

Please check the examination details below before entering your candidate information

Candidate

surname:

USMAN

Centre Number

Candidate Number

Other names:

Grade A* Scholar

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Pearson Edexcel International Advanced Level

Chemistry

International Advanced Level

UNIT 4: Rates, Equilibria and Further Organic Chemistry

Paper Reference: WCH14/01/W2-REV

Time Allowed: 2 hours 30 minutes

Topic Focus: 11A.3 Determining Orders & 11A.4 Reaction

Mechanisms

Total Marks: 150 Marks

INSTRUCTIONS

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page with your name, centre number and candidate number.
- Answer all questions.
- Answer the questions in the spaces provided — there may be more space than you need.
- Show all the steps in any calculations and state the units of numerical values.
- Calculators may be used.

INFORMATION

- The total mark for this paper is 150.
- The marks for each question are shown in brackets — use this as a guide as to how much time to spend on each question.
- In questions marked with an asterisk (*), marks will be awarded for your ability to structure your answer logically, showing how the points that you make are related or follow on from each other.
- A Periodic Table is printed on the back page of this question paper.

ADVICE

- Read each question carefully before you start to answer it.
- Keep an eye on the time.
- Try to answer every question.
- Check your answers if you have time at the end.

MENTORA ACADEMY • **WEEK 2 KINETICS & MECHANISMS MASTERY EXAM**

Prepared exclusively for candidate USMAN · Strict timed simulation conditions

SECTION A

Answer ALL questions in this section.

You should spend no more than 40 minutes on this section.

Use the spaces provided in this question paper.

For each question, select one answer from A to D and put a cross in the box [X].

If you change your mind, put a line through the box and then mark your new answer with a cross.

1 The decomposition of hydrogen peroxide, $2\text{H}_2\text{O}_2(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$, is first order with respect to H_2O_2 . The half-life of H_2O_2 is 240 s. What fraction of the original H_2O_2 remains after 720 s?

- ☐ A $1/3$
☐ B $1/4$
☐ C $1/8$
☐ D $1/16$

(Total for Question 1 = 1 mark)

2 For a second-order reaction with respect to reactant A, what happens to the half-life ($t_{1/2}$) if the initial concentration of A is tripled?

- ☐ A The half-life triples ($3x$).
☐ B The half-life decreases to one-third ($1/3x$).
☐ C The half-life decreases to one-ninth ($1/9x$).
☐ D The half-life remains unchanged.

(Total for Question 2 = 1 mark)

3 Which of the following graphs produces a straight line for a first-order reaction with respect to reactant X?

- ☐ A $[X]$ against time t
☐ B $\ln[X]$ against time t
☐ C $1/[X]$ against time t
☐ D $1/[X]^2$ against time t

(Total for Question 3 = 1 mark)

4 For a reaction with rate = $k[A]^0$, how does the half-life vary with the initial concentration $[A]_0$?

- ☐ A $t_{1/2}$ is directly proportional to $[A]_0$.
- ☐ B $t_{1/2}$ is inversely proportional to $[A]_0$.
- ☐ C $t_{1/2}$ is independent of $[A]_0$.
- ☐ D $t_{1/2}$ is proportional to $[A]_0^2$.

(Total for Question 4 = 1 mark)

5 A plot of $1/[A]$ against time t for a reaction gives a straight line with a positive slope equal to $0.084 \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1}$. What is the overall order of the reaction with respect to A?

- ☐ A Zero order
- ☐ B First order
- ☐ C Second order
- ☐ D Third order

(Total for Question 5 = 1 mark)

6 The reaction between 2-bromo-2-methylpropane and aqueous sodium hydroxide follows the rate law rate = $k[(\text{CH}_3)_3\text{CBr}]$. What is the rate-determining step?

- ☐ A Nucleophilic attack of OH^- on the central carbon atom.
- ☐ B Heterolytic fission of the C-Br bond to form a carbocation intermediate.
- ☐ C Proton transfer from the alcohol to form water.
- ☐ D Simultaneous bond making and bond breaking in a pentacoordinate transition state.

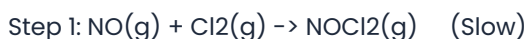
(Total for Question 6 = 1 mark)

7 When (R)-2-bromobutane reacts with aqueous hydroxide ions via an SN2 mechanism, what is the stereochemical outcome of the product, butan-2-ol?

- ☐ A A 50:50 racemic mixture of (R) and (S) enantiomers.
- ☐ B 100% (S)-butan-2-ol due to complete Walden inversion of configuration.
- ☐ C 100% (R)-butan-2-ol with full retention of configuration.
- ☐ D An optically inactive meso compound.

(Total for Question 7 = 1 mark)

8 Consider the reaction mechanism:

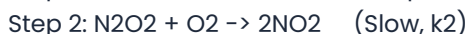


What is the expected rate equation for this mechanism?

- ☐ A rate = $k[\text{NO}]^2 [\text{Cl}_2]$
- ☐ B rate = $k[\text{NO}][\text{Cl}_2]$
- ☐ C rate = $k[\text{NOCl}_2][\text{NO}]$
- ☐ D rate = $k[\text{NO}]^2$

(Total for Question 8 = 1 mark)

9 The reaction $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$ follows the rate law rate = $k[\text{NO}]^2 [\text{O}_2]$. A proposed mechanism is:



What is the molecularity of the rate-determining step (Step 2)?

- ☐ A Unimolecular
- ☐ B Bimolecular
- ☐ C Termolecular
- ☐ D Zero-molecular

(Total for Question 9 = 1 mark)

10 In a reaction profile (energy vs reaction coordinate), what feature represents an intermediate in a multi-step mechanism?

- ☐ A A potential energy maximum.
- ☐ B A local potential energy minimum between two transition states.
- ☐ C The difference in energy between reactants and products.
- ☐ D The highest point on the curve.

(Total for Question 10 = 1 mark)

11 A reaction has the rate equation $\text{rate} = k[\text{P}][\text{Q}]$. In which case could this rate equation be consistent with a multi-step mechanism?

- ☐ A P and Q collide in the slow first step of the mechanism.
- ☐ B P decomposes in a slow first step and Q is added in a fast second step.
- ☐ C P and Q are both involved in a fast step after the rate-determining step.
- ☐ D Two molecules of P react in the rate-determining step.

(Total for Question 11 = 1 mark)

12 Why are elementary reaction steps involving termolecular collisions (three species colliding simultaneously) exceedingly rare in chemistry?

- ☐ A Termolecular collisions are strictly forbidden by thermodynamics.
- ☐ B The probability of three independent particles colliding simultaneously with correct orientation and sufficient energy is extremely low.
- ☐ C Termolecular collisions only occur in solids.
- ☐ D Termolecular steps always have zero activation energy.

(Total for Question 12 = 1 mark)

13 A student determines the initial rate of reaction from a concentration–time curve by drawing a tangent. At which point must the tangent be constructed to obtain the initial rate?

- ☐ A At $t = t_{1/2}$
- ☐ B At $t = 0$
- ☐ C At the point of inflection
- ☐ D At the completion of the reaction

(Total for Question 13 = 1 mark)

14 For a reaction $A \rightarrow \text{products}$, the time taken for $[A]$ to fall from 0.80 mol dm^{-3} to 0.40 mol dm^{-3} is 50 s. The time taken for $[A]$ to fall from 0.40 to 0.20 mol dm^{-3} is also 50 s. What is the value of the rate constant k ?

- ☐ A 0.0139 s^{-1}
- ☐ B 0.0080 s^{-1}
- ☐ C 0.0200 s^{-1}
- ☐ D 0.693 s^{-1}

(Total for Question 14 = 1 mark)

15 Consider the reaction between propanone and bromine in acid: $\text{CH}_3\text{COCH}_3 + \text{Br}_2 \rightarrow \text{CH}_3\text{COCH}_2\text{Br} + \text{H}^+ + \text{Br}^-$. The rate equation is $\text{rate} = k[\text{CH}_3\text{COCH}_3][\text{H}^+]$. What is the order of reaction with respect to Br_2 ?

- ☐ A 0
- ☐ B 1
- ☐ C 2
- ☐ D -1

(Total for Question 15 = 1 mark)

16 A reaction between species X and Y is investigated. When $[X]$ is kept constant, a plot of rate against $[Y]$ gives a horizontal straight line. What is the order with respect to Y?

- ☐ A 0
- ☐ B 1
- ☐ C 2
- ☐ D 0.5

(Total for Question 16 = 1 mark)

17 When $\log_{10}(\text{rate})$ is plotted against $\log_{10}[\text{reactant}]$, what information does the gradient of the straight line provide?

- ☐ A The rate constant k
- ☐ B The activation energy E_a
- ☐ C The order of reaction with respect to that reactant
- ☐ D The half-life of the reaction

(Total for Question 17 = 1 mark)

18 For a second-order reaction $\text{rate} = k[A]^2$ with $k = 0.25 \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ and $[A]_0 = 0.50 \text{ mol dm}^{-3}$, what is the initial half-life?

- ☐ A 2.0 s
- ☐ B 4.0 s
- ☐ C 8.0 s
- ☐ D 1.39 s

(Total for Question 18 = 1 mark)

19 In an S_N1 mechanism for the hydrolysis of a tertiary bromoalkane, what is the geometry of the carbocation intermediate?

- ☐ A Trigonal bipyramidal
- ☐ B Trigonal planar with bond angles of 120°
- ☐ C Tetrahedral with bond angles of 109.5°
- ☐ D Pyramidal with bond angles of 107°

(Total for Question 19 = 1 mark)

20 The reaction $2\text{NO}_2(\text{g}) + \text{F}_2(\text{g}) \rightarrow 2\text{NO}_2\text{F}(\text{g})$ has the experimental rate law $\text{rate} = k[\text{NO}_2][\text{F}_2]$. Which of the following mechanisms is consistent with this rate law?

- ☐ A $\text{NO}_2 + \text{F}_2 \rightarrow \text{NO}_2\text{F} + \text{F}$ (Slow); $\text{NO}_2 + \text{F} \rightarrow \text{NO}_2\text{F}$ (Fast)
- ☐ B $2\text{NO}_2 \rightarrow \text{N}_2\text{O}_4$ (Fast); $\text{N}_2\text{O}_4 + \text{F}_2 \rightarrow 2\text{NO}_2\text{F}$ (Slow)
- ☐ C $2\text{NO}_2 + \text{F}_2 \rightarrow 2\text{NO}_2\text{F}$ (One-step termolecular)
- ☐ D $\text{F}_2 \rightarrow 2\text{F}$ (Slow); $\text{NO}_2 + \text{F} \rightarrow \text{NO}_2\text{F}$ (Fast)

(Total for Question 20 = 1 mark)

21 Why does an S_N1 substitution reaction at a chiral carbon centre typically result in a racemic mixture?

- ☐ A The nucleophile can only attack from the back face.
- ☐ B The planar carbocation intermediate can be attacked by the nucleophile with equal probability from either face.
- ☐ C The leaving group shields both faces equally.
- ☐ D The carbocation undergoes rapid tautomerisation.

(Total for Question 21 = 1 mark)

22 For the reaction $A + B \rightarrow C$, the rate equation is $\text{rate} = k[A]^2$. If the initial concentration of B is doubled, what happens to the initial rate of reaction?

- ☐ A Doubles (2x)
☐ B Quadruples (4x)
☐ C Halves (0.5x)
☐ D Remains unchanged

(Total for Question 22 = 1 mark)

23 A reaction has a rate equation $\text{rate} = k[A][B]^2$. In an experiment, [A] is doubled and [B] is halved. By what factor does the rate change?

- ☐ A Multiplied by 2
☐ B Multiplied by 0.5 (halved)
☐ C Multiplied by 4
☐ D Unchanged

(Total for Question 23 = 1 mark)

24 For a first-order reaction $A \rightarrow B$, the initial rate is $0.040 \text{ mol dm}^{-3} \text{ s}^{-1}$ when $[A] = 0.20 \text{ mol dm}^{-3}$. What is the rate of reaction when [A] has decreased to $0.050 \text{ mol dm}^{-3}$?

- ☐ A $0.010 \text{ mol dm}^{-3} \text{ s}^{-1}$
☐ B $0.020 \text{ mol dm}^{-3} \text{ s}^{-1}$
☐ C $0.005 \text{ mol dm}^{-3} \text{ s}^{-1}$
☐ D $0.040 \text{ mol dm}^{-3} \text{ s}^{-1}$

(Total for Question 24 = 1 mark)

25 In the mechanism:

Step 1: $A + B \rightleftharpoons C$ (Fast equilibrium, $K_c = [C]/([A][B])$)

Step 2: $C + D \rightarrow E$ (Slow, k_2)

What is the overall rate equation in terms of the original reactants?

- ☐ A rate = $k_2[C][D]$
- ☐ B rate = $k[A][B][D]$
- ☐ C rate = $k[A][B]$
- ☐ D rate = $k[C]$

(Total for Question 25 = 1 mark)

26 A student measures the rate of reaction by drawing a tangent to a curve of $[A]$ vs t . If the tangent at $t = 20$ s has $\Delta[A] = -0.15 \text{ mol dm}^{-3}$ over $\Delta t = 60$ s, what is the rate of reaction at $t = 20$ s?

- ☐ A $0.0025 \text{ mol dm}^{-3} \text{ s}^{-1}$
- ☐ B $0.0050 \text{ mol dm}^{-3} \text{ s}^{-1}$
- ☐ C $0.0075 \text{ mol dm}^{-3} \text{ s}^{-1}$
- ☐ D $0.15 \text{ mol dm}^{-3} \text{ s}^{-1}$

(Total for Question 26 = 1 mark)

27 Which statement about the transition state of an elementary reaction is correct?

- ☐ A It can be isolated and bottled in the laboratory.
- ☐ B It has a lower potential energy than both reactants and products.
- ☐ C It is a transient arrangement of atoms at maximum potential energy with partially formed and partially broken bonds.
- ☐ D It is identical in structure to a reaction intermediate.

(Total for Question 27 = 1 mark)

28 The reaction $2A + B \rightarrow C$ follows the rate law $\text{rate} = k[A]^2[B]$. Which proposed mechanism is INCONSISTENT with this rate law?

- ☐ A $A + A \rightleftharpoons A_2$ (Fast); $A_2 + B \rightarrow C$ (Slow)
- ☐ B $A + B \rightleftharpoons AB$ (Fast); $AB + A \rightarrow C$ (Slow)
- ☐ C $A + B \rightarrow AB$ (Slow); $AB + A \rightarrow C$ (Fast)
- ☐ D $2A + B \rightarrow C$ (Single termolecular step)

(Total for Question 28 = 1 mark)

29 For the gas-phase reaction $H_2 + I_2 \rightarrow 2HI$, the reaction is first order in H_2 and first order in I_2 . What happens to the rate if the pressure of the system is tripled at constant temperature?

- ☐ A Increases by 3x
- ☐ B Increases by 6x
- ☐ C Increases by 9x
- ☐ D Remains unchanged

(Total for Question 29 = 1 mark)

30 A first-order reaction has $k = 3.5 \times 10^{-4} \text{ s}^{-1}$. How long will it take for the concentration of the reactant to decrease to 25% of its initial value?

- ☐ A 990 s
- ☐ B 1980 s
- ☐ C 3960 s
- ☐ D 2857 s

(Total for Question 30 = 1 mark)

SECTION B

Answer ALL questions. Write your answers in the spaces provided.

31 Dinitrogen pentoxide decomposes in tetrachloromethane solvent according to the stoichiometric equation:



The concentration of N_2O_5 was measured at 318 K over time. The experimental data are shown below:

Time t / s	0	400	800	1200	1600	2000	2400	2800
	1.000	0.669	0.448	0.300	0.201	0.134	0.090	0.060

(a) (i) On the axes below, sketch the concentration–time curve for the decomposition of N_2O_5 . (2)



(ii) Show that the reaction is first-order with respect to N_2O_5 by determining two successive half-lives from the data in the table. (4)

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First half-life, $t_{1/2}(1) = \dots\dots\dots \text{ s}$

Second half-life, $t_{1/2}(2) = \dots\dots\dots \text{ s}$

- (b) A student draws a tangent to the curve at $t = 0$ s. The tangent intersects the concentration axis at $1.000 \text{ mol dm}^{-3}$ and the time axis at 1440 s. Calculate the initial rate of reaction in $\text{mol dm}^{-3} \text{ s}^{-1}$. (3)

Initial rate = $\text{mol dm}^{-3} \text{ s}^{-1}$

- (c) Using your value for the half-life from (a)(ii), calculate the rate constant, k , at 318 K. Include units in your answer. (3)

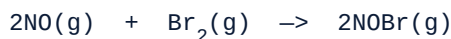
$k =$ units

- (d) Explain why the stoichiometric coefficient of N_2O_5 in the balanced equation is 2, yet the reaction is only first order with respect to N_2O_5 . (3)

- (e) State the mathematical expression for the gradient of a plot of $\ln[\text{N}_2\text{O}_5]$ against time t , and state whether this gradient is positive or negative. (3)

(Total for Question 31 = 18 marks)

32 Nitrogen monoxide reacts with bromine vapour at 273 K according to the equation:



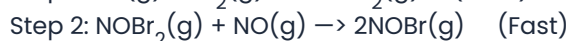
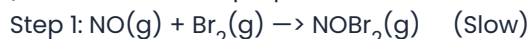
Kinetic experiments demonstrate that the reaction is second order with respect to NO and first order with respect to Br_2 :

$$\text{Rate} = k [\text{NO}]^2 [\text{Br}_2].$$

(a) A chemist suggests that the reaction occurs via a single termolecular collision between two molecules of NO and one molecule of Br_2 . Explain why termolecular elementary steps are extremely improbable in gas-phase kinetics.

(3)

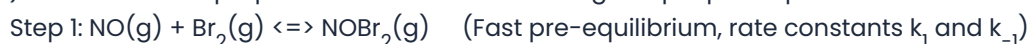
(b) Another chemist proposes Mechanism A:



Explain why Mechanism A is completely inconsistent with the experimentally determined rate equation.

(3)

(c) A third chemist proposes Mechanism B involving a rapid pre-equilibrium:



Derive the overall rate equation from Mechanism B. Show clearly how the concentration of the intermediate NOBr_2 is expressed in terms of the reactants, and relate the observed rate constant k to the elementary constants k_1 , k_{-1} and k_2 .

(6)

(d) State the molecularity of Step 1 in the forward direction and the molecularity of Step 2 in Mechanism B.

(2)

Molecularity of Step 1 =

Molecularity of Step 2 =

(e) In an experiment at 273 K, the initial concentrations were $[\text{NO}] = 2.50 \times 10^{-3} \text{ mol dm}^{-3}$ and $[\text{Br}_2] = 1.20 \times 10^{-3} \text{ mol dm}^{-3}$. The initial rate of formation of NOBr was $1.80 \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1}$.

Calculate the rate constant, k , for the consumption of bromine, and state its units.

(4)

$k = \dots\dots\dots$ units $\dots\dots\dots$

(Total for Question 32 = 18 marks)

33 Halogenoalkanes undergo nucleophilic substitution with aqueous hydroxide ions by two distinct mechanisms, S_N1 and S_N2 .

(a) 1-bromobutane reacts with aqueous sodium hydroxide to form butan-1-ol via an S_N2 mechanism.

(i) Write the rate equation for this reaction.

(ii) Describe the mechanism of this reaction. In your answer, refer to the attacking nucleophile, the transition state, bond formation/breaking, and explain why the mechanism is classified as 'bimolecular'.

(6)

Rate equation =

(b) 2-bromo-2-methylpropane reacts with aqueous sodium hydroxide to form 2-methylpropan-2-ol via an S_N1 mechanism.

(i) Write the rate equation for this reaction.

(ii) Outline the two elementary steps of this mechanism and identify the rate-determining step.

(4)

Rate equation =

(c) Optically active (R)-2-bromobutane is hydrolysed by warm aqueous sodium hydroxide. Predict and explain the stereochemical outcome of the product butan-2-ol if the reaction proceeds entirely by an S_N2 mechanism. Contrast this with the stereochemical outcome if the reaction proceeded via an S_N1 mechanism.

(5)

(d) In two separate experiments, the concentration of aqueous sodium hydroxide is doubled while the concentration of the halogenoalkane is kept constant.

State and explain the effect of doubling $[\text{OH}^-]$ on:

(i) The rate of hydrolysis of 1-bromobutane (S_N2).

(ii) The rate of hydrolysis of 2-bromo-2-methylpropane (S_N1).

(3)**(Total for Question 33 = 18 marks)**

34 Phenolphthalein (P^{2-}) is pink in alkaline solution. In strongly alkaline conditions ($[\text{OH}^-] > 0.1 \text{ mol dm}^{-3}$), it reacts slowly with hydroxide ions to form a colourless carbinol compound (POH^{3-}):



$$\text{Rate} = k [\text{P}^{2-}]^m [\text{OH}^-]^n.$$

*(a) Extended Writing: A chemist wishes to determine the order of reaction, m , with respect to phenolphthalein by graphical analysis of continuous colorimetric data.

Compare and evaluate three distinct graphical methods for determining the order m from concentration-time data:

Method 1: Determining rates from tangents at various concentrations and plotting rate vs $[P^{2-}]$.

Method 2: Measuring successive half-lives from the concentration-time curve.

Method 3: Testing integrated rate law plots ($\ln[P^{2-}]$ vs t and $1/[P^{2-}]$ vs t).

Discuss the advantages, apparatus limitations, and precision of each method.

(6)

(b) In Experiment 1, $[\text{OH}^-]$ is kept in large excess at $0.500 \text{ mol dm}^{-3}$ while $[\text{P}^{2-}]_0 = 5.00 \times 10^{-5} \text{ mol dm}^{-3}$. The following absorbance data were recorded:

Time t / s	0	60	120	180	240	300
Absorbance, A	1.200	0.849	0.600	0.424	0.300	0.212
$\ln A$	0.182	-0.164	-0.511	-0.858	-1.204	-1.551

(i) Using the half-life method, show that the reaction is first-order with respect to phenolphthalein ($m = 1$).

(ii) Calculate the pseudo-first-order rate constant, k' , from the gradient of the $\ln A$ vs t data. **(6)**

Half-life = s

$k' = \dots\dots\dots \text{s}^{-1}$

(c) The experiment was repeated at four different concentrations of sodium hydroxide. The apparent pseudo-rate constants k' are tabulated below:

$[\text{OH}^-] / \text{mol dm}^{-3}$	0.100	0.200	0.400	0.800
	1.15	2.31	4.62	9.23

Deduce the order of reaction, n , with respect to hydroxide ions. Write the overall rate equation and calculate the true rate constant k with units. **(6)**

Order with respect to OH^- =

Rate equation =

k = units

(Total for Question 34 = 18 marks)

35 The decomposition of ozone into oxygen occurs in the stratosphere according to the equation:



Kinetic investigations establish that the reaction has the unusual empirical rate equation:

$$\text{Rate} = k [\text{O}_3]^2 [\text{O}_2]^{-1}.$$

(a) (i) State the overall order of the reaction.

(ii) Explain the physical meaning of a negative order of reaction with respect to oxygen.

(3)

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Overall order =

(b) The following two-step mechanism is proposed:

Step 1: $\text{O}_3(\text{g}) \rightleftharpoons \text{O}_2(\text{g}) + \text{O}(\text{g})$ (Fast equilibrium, forward k_f , reverse k_{-f})

Step 2: $\text{O}_3(\text{g}) + \text{O}(\text{g}) \rightarrow 2\text{O}_2(\text{g})$ (Slow, rate constant k_2)

Derive the rate equation from this mechanism. Show that it is identical in form to the empirical rate equation, and express the observed rate constant k in terms of k_f , k_{-f} and k_2 .

(6)

(c) At a certain altitude in the stratosphere, $[\text{O}_3] = 4.00 \times 10^{-8} \text{ mol dm}^{-3}$ and $[\text{O}_2] = 8.00 \times 10^{-6} \text{ mol dm}^{-3}$. The rate constant k at this temperature is $5.00 \times 10^{11} \text{ s}^{-1}$.

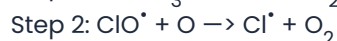
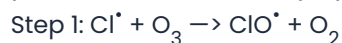
(i) Calculate the rate of decomposition of ozone under these conditions in $\text{mol dm}^{-3} \text{ s}^{-1}$.

(ii) Calculate the rate of formation of oxygen, $d[\text{O}_2]/dt$.

(4)

Rate of decomposition of $\text{O}_3 = \dots\dots\dots \text{mol dm}^{-3} \text{ s}^{-1}$
 $d[\text{O}_2]/dt = \dots\dots\dots \text{mol dm}^{-3} \text{ s}^{-1}$

(d) Chlorine free radicals ($\text{Cl}^\bullet$) catalytically destroy ozone via a two-step cycle:



(i) Explain how this catalytic cycle provides an alternative reaction pathway with a lower activation energy.

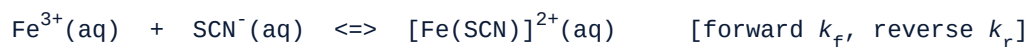
(ii) Why does the chlorine radical not appear in the overall chemical equation?

(5)**(Total for Question 35 = 18 marks)**

SECTION C

Answer ALL questions. Write your answers in the spaces provided.

36 The fast complexation reaction between iron(III) ions and thiocyanate ions in acidic solution produces an intensely red complex:



Because the reaction is complete in less than 200 milliseconds, it is studied using the stopped-flow spectrophotometric technique at 450 nm.

(a) Describe the principles of the stopped-flow technique. Explain how rapid mixing and the stopping syringe allow reactions occurring on the millisecond timescale to be measured spectrophotometrically.

(4)

(b) In a series of stopped-flow runs at 298 K, $[\text{SCN}^{-}]_0 = 1.00 \times 10^{-4} \text{ mol dm}^{-3}$ and $[\text{Fe}^{3+}]$ was present in large excess. The measured pseudo-first-order rate constants k_{obs} are given below:

$[\text{Fe}^{3+}] / 10^{-2} \text{ mol dm}^{-3}$	1.00	2.00	4.00	6.00
	6.10	10.80	20.20	29.60

For this reversible reaction under pseudo-first-order conditions:

$$k_{\text{obs}} = k_f [\text{Fe}^{3+}] + k_r$$

(i) Determine the value of the forward rate constant, k_f , from the gradient of k_{obs} against $[\text{Fe}^{3+}]$, including units.

(ii) Determine the reverse rate constant, k_r , from the y-intercept, including units.

(iii) Calculate the equilibrium constant, $K_c = k_f / k_r$, for the complexation.

(8)

$k_f =$ units

$k_r =$ units

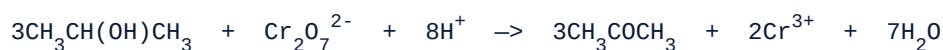
$K_c =$ units

(c) State what optical wavelength filter should be chosen for following the concentration of $[\text{Fe}(\text{SCN})]^{2+}$ by colorimetry, and justify your choice with reference to complementary colours.

(3)

(Total for Question 36 = 15 marks)

37 Propan-2-ol is oxidised to propanone by acidified potassium dichromate(VI) in aqueous solution:



The kinetics of this redox reaction were studied at 298 K using continuous colorimetry.

(a) Explain how colorimetry can monitor the progress of this reaction. State the colour change observed and specify which absorbing species can be followed.

(3)

(b) The following initial rate data were obtained at 298 K:

Exp	[propan-2-ol] / M	[Cr ₂ O ₇ ²⁻] / M	[H ⁺] / M	Initial Rate / 10 ⁻⁵ mol dm ⁻³ s ⁻¹
1	0.200	0.0100	0.500	1.60
2	0.400	0.0100	0.500	3.20
3	0.200	0.0200	0.500	3.20
4	0.200	0.0100	1.000	6.40

Deduce the order of reaction with respect to propan-2-ol, dichromate(VI), and hydrogen ions. Write the rate equation and calculate the rate constant k with units.

(8)

Order with respect to propan-2-ol =

Order with respect to Cr₂O₇²⁻ =

Order with respect to H⁺ =

Rate equation =

k = units

(c) When the reaction is carried out with deuterated propan-2-ol, $(\text{CH}_3)_2\text{CD}(\text{OH})$, the initial rate decreases by a factor of 6.7 (primary kinetic isotope effect).

Explain what this observation reveals about the bond broken in the rate-determining step of the mechanism.

(4)

(Total for Question 37 = 15 marks)

TOTAL FOR SECTION C = 30 MARKS

TOTAL FOR PAPER = 150 MARKS

**END OF QUESTION
PAPER**

DATA SHEET – THE PERIODIC TABLE OF THE ELEMENTS

Group 1	Group 2	Group 3	Group 4	Group 5	Group 6	Group 7	Group 0
1 H 1.0	-	-	-	-	-	-	2 He 4.0
3 Li 6.9	4 Be 9.0	5 B 10.8	6 C 12.0	7 N 14.0	8 O 16.0	9 F 19.0	10 Ne 20.2
11 Na 23.0	12 Mg 24.3	13 Al 27.0	14 Si 28.1	15 P 31.0	16 S 32.1	17 Cl 35.5	18 Ar 39.9
19 K 39.1	20 Ca 40.1	31 Ga 69.7	32 Ge 72.6	33 As 74.9	34 Se 79.0	35 Br 79.9	36 Kr 83.8
37 Rb 85.5	38 Sr 87.6	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
55 Cs 132.9	56 Ba 137.3	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po [209]	85 At [210]	86 Rn [222]

Physical Constant / Formula	Symbol / Expression	Value & Units
Gas constant	R	8.314 J mol ⁻¹ K ⁻¹
Molar volume of gas at r.t.p.	V _m	24.0 dm ³ mol ⁻¹ (24,000 cm ³ mol ⁻¹)
First-order half-life relationship	t _{1/2}	t _{1/2} = ln 2 / k = 0.693 / k
Second-order half-life relationship	t _{1/2}	t _{1/2} = 1 / (k [A] ₀)
Integrated first-order rate equation	ln[A]	ln[A] _t = -k t + ln[A] ₀
Integrated second-order rate equation	1/[A]	1/[A] _t = k t + 1/[A] ₀

PEARSON EDEXCEL INTERNATIONAL ADVANCED LEVEL

CHEMISTRY UNIT 4 (WCH14/01) – ADVANCED MASTERY QUIZ

CONFIDENTIAL TEACHER'S MARK SCHEME & EXAMINER GUIDANCE (150 MARKS)

SECTION A ANSWER KEY & EXPLANATIONS (30 MARKS)

Q1	Key: C	720 s represents exactly 3 half-lives ($720 / 240 = 3$). Fraction remaining = $(1/2)^3 = 1/8$. (1)
Q2	Key: B	For second order, $t_{1/2} = 1 / (k[A]_0)$. Tripling initial concentration divides the half-life by 3 ($1/3x$). (1)
Q3	Key: B	Integrated first-order rate law: $\ln[X] = -kt + \ln[X]_0$; a plot of $\ln[X]$ vs t yields a straight line with slope $-k$. (1)
Q4	Key: A	For zero order, rate = $-d[A]/dt = k \Rightarrow [A]_t = -kt + [A]_0$. At half-life, $[A] = [A]_0/2 \Rightarrow t_{1/2} = [A]_0 / (2k)$. Directly proportional to $[A]_0$. (1)
Q5	Key: C	Integrated second order: $1/[A] = kt + 1/[A]_0$; linear $1/[A]$ vs t plot confirms second order with slope = k . (1)
Q6	Key: B	S_N1 is unimolecular; the rate-determining step is slow heterolytic bond fission of C-Br forming a stable tertiary carbocation. (1)
Q7	Key: B	S_N2 occurs via backside attack of OH^- opposite the departing Br^- , causing complete inversion of configuration (Walden inversion). (1)
Q8	Key: B	Step 1 is the slow rate-determining step, so rate depends only on the reactants in Step 1: rate = $k[NO][Cl_2]$. (1)
Q9	Key: B	Step 2 involves the collision of two particles (N_2O_2 and O_2), so its molecularity is bimolecular (molecularity = 2). (1)
Q10	Key: B	An intermediate is a real chemical species located at a local potential energy minimum between two transition state activation energy peaks. (1)
Q11	Key: A	If rate = $k[P][Q]$, the rate-determining step must involve one molecule of P and one molecule of Q colliding. (1)
Q12	Key: B	The simultaneous collision of three particles with proper orientation and sufficient kinetic energy has negligible statistical probability. (1)
Q13	Key: B	Initial rate is determined by drawing a tangent at time $t = 0$ s. (1)
Q14	Key: A	Constant half-life of 50 s confirms first-order kinetics. $k = \ln 2 / t_{1/2} = 0.693 / 50 = 0.0139 \text{ s}^{-1}$. (1)
Q15	Key: A	Bromine does not appear in the rate equation, so the reaction is zero order with respect to Br_2 (involved after the RDS). (1)
Q16	Key: A	A horizontal rate vs $[Y]$ line means rate does not change when $[Y]$ varies $\Rightarrow$ zero order in Y. (1)
Q17	Key: C	$\log_{10}(\text{rate}) = \log_{10}(k) + n \log_{10}[\text{reactant}]$; gradient = n (the order of reaction). (1)
Q18	Key: C	For second order, $t_{1/2} = 1 / (k[A]_0) = 1 / (0.25 \times 0.50) = 1 / 0.125 = 8.0$ s. (1)
Q19	Key: B	A carbocation has 3 bonding pairs and zero lone pairs on carbon (sp^2 hybridised), adopting a trigonal planar geometry with 120° bond angles. (1)
Q20	Key: A	In mechanism A, the slow step is bimolecular with one NO_2 and one F_2 , giving rate = $k[NO_2][F_2]$, matching experimental observation. (1)
Q21	Key: B	The intermediate carbocation is planar; the nucleophile has an equal 50% probability of attacking from the front or back face, yielding a racemate. (1)
Q22	Key: D	Rate equation is rate = $k[A]^2$; reactant B is zero order, so doubling $[B]$ has no effect on the rate. (1)

Q23	Key: B	New rate proportional to $[2A] * [0.5 B]^2 = 2 * 0.25 = 0.5$ (the initial rate halves). (1)
Q24	Key: A	For first order, rate is directly proportional to $[A]$. When $[A]$ drops by 4x ($0.20 \rightarrow 0.050$), rate drops by 4x: $0.040 / 4 = 0.010 \text{ mol dm}^{-3} \text{ s}^{-1}$. (1)
Q25	Key: B	From pre-equilibrium, $[C] = K_c[A][B]$. Rate = $k_2[C][D] = k_2 K_c [A][B][D] = k[A][B][D]$. (1)
Q26	Key: A	Rate = $-\text{gradient} = -(-0.15 \text{ mol dm}^{-3} / 60 \text{ s}) = +0.0025 \text{ mol dm}^{-3} \text{ s}^{-1}$. (1)
Q27	Key: C	A transition state is an activated complex at maximum potential energy with partial bonds that cannot be isolated. (1)
Q28	Key: C	If Step 1 ($A + B \rightarrow AB$) were slow, the rate law would be rate = $k[A][B]$, which contradicts the experimental second order in A. (1)
Q29	Key: C	Tripling pressure triples all gas concentrations: new rate = $k(3[H_2])(3[I_2]) = 9 \times$ original rate. (1)
Q30	Key: C	For $[A]$ to fall to 25% requires two half-lives ($100\% \rightarrow 50\% \rightarrow 25\%$). $t_{1/2} = \ln 2 / (3.5 \times 10^{-4}) = 1980 \text{ s}$. Two half-lives = $2 * 1980 = 3960 \text{ s}$. (1)

QUESTION 31: Graphical Kinetics of N₂O₅ Decomposition (18 Marks)

(a) Graph Sketch & Half-Lives	(i) Smooth curve starting at (0, 1.000) showing exponential decay approaching zero asymptotically (2). (ii) First half-life: $[N_2O_5]$ falls from 1.000 to 0.500 mol dm ⁻³ at t approx 690 s ($t_{1/2}(1) = 690 \text{ s}$) (2). Second half-life: falls from 0.500 to 0.250 mol dm ⁻³ at t approx 1380 s ($t_{1/2}(2) = 1380 - 690 = 690 \text{ s}$) (1). Successive half-lives are constant within experimental error, confirming first-order kinetics (1).	[6]
(b) Initial Rate from Tangent	Initial rate = $-\text{gradient at } t = 0 \text{ s} = -(0 - 1.000) / (1440 - 0)$ (2). $= 6.94 \times 10^{-4} \text{ mol dm}^{-3} \text{ s}^{-1}$ (1).	[3]
(c) Rate Constant k	$k = \ln 2 / t_{1/2} = 0.693 / 690 \text{ s} = 1.00 \times 10^{-3}$ (or 1.01×10^{-3}) (2). Units: s ⁻¹ (1).	[3]
(d) Stoichiometry vs Order	Reaction is multi-step; the rate-determining step involves only one molecule of N ₂ O ₅ undergoing unimolecular fission ($N_2O_5 \rightarrow NO_2 + NO_3$) (2). Stoichiometric coefficients represent overall mole balance and do not indicate reaction order or mechanism (1).	[3]
(e) ln[N₂O₅] Plot Gradient	Integrated first-order law: $\ln[N_2O_5]_t = -k t + \ln[N_2O_5]_0$ (1). Gradient = $-k$ (1). Gradient is negative because reactant concentration decreases over time (1).	[3]

QUESTION 32: Rate Laws and Reaction Mechanisms (18 Marks)

(a) Improbability of Termolecular Step	Simultaneous collision of three particles requires all three to be in the same spatial volume at the same instant (1). All three particles must have kinetic energy $\geq E_a$ and correct molecular collision orientations (1). Statistical probability of this is negligible compared to successive bimolecular collisions (1).	[3]
(b) Inconsistency of Mechanism A	If Step 1 were rate-determining, the rate equation would be rate = $k[NO][Br_2]$ (2). This is first order in NO, which directly contradicts the experimental second-order dependence in NO (1).	[3]

(c) Derivation from Mechanism B	Step 2 is the slow RDS: $\text{Rate} = k_2 [\text{NOBr}_2][\text{NO}]$ (1). From fast pre-equilibrium Step 1: $\text{rate}_{\text{forward}} = \text{rate}_{\text{reverse}} \Rightarrow k_1 [\text{NO}][\text{Br}_2] = k_{-1} [\text{NOBr}_2]$ (2). Express intermediate: $[\text{NOBr}_2] = (k_1 / k_{-1}) [\text{NO}][\text{Br}_2] = K_c [\text{NO}][\text{Br}_2]$ (1). Substitute into rate equation: $\text{Rate} = k_2 (k_1 / k_{-1}) [\text{NO}][\text{Br}_2] [\text{NO}] = (k_2 k_1 / k_{-1}) [\text{NO}]^2 [\text{Br}_2]$ (1). Matches experimental rate law with $k = (k_2 k_1 / k_{-1})$ (1).	[6]
(d) Molecularity of Steps	Step 1 forward molecularity: Bimolecular (two molecules: NO and Br₂) (1). Step 2 molecularity: Bimolecular (two molecules: NOBr₂ and NO) (1).	[2]
(e) Calculation of Rate Constant	Rate of consumption of Br₂ = 0.5 * rate of formation of NOBr = 0.5 * (1.80 x 10⁻⁵) = 9.00 x 10⁻⁶ mol dm⁻³ s⁻¹ (1). $k = \text{Rate} / ([\text{NO}]^2 [\text{Br}_2]) = 9.00 \times 10^{-6} / ((2.50 \times 10^{-3})^2 * (1.20 \times 10^{-3}))$ (1). = 9.00 x 10⁻⁶ / 7.50 x 10⁻⁹ = 1.20 x 10³ (or 1200) (1). Units: dm⁶ mol⁻² s⁻¹ (1).	[4]

QUESTION 33: Kinetics & Stereochemistry of Halogenoalkanes (18 Marks)

(a) SN2 Mechanism & Rate	(i) $\text{Rate} = k[1\text{-bromobutane}][\text{OH}^-]$ (1). (ii) OH⁻ nucleophile attacks carbon bonded to Br from the back (opposite to Br) (1). Forms a single pentacoordinate transition state with partially formed C-OH bond and partially broken C-Br bond (2). Classified as bimolecular because two species (halogenoalkane and hydroxide) are involved in the single rate-determining step (2).	[6]
(b) SN1 Mechanism & Rate	(i) $\text{Rate} = k[2\text{-bromo-2-methylpropane}]$ (1). (ii) Step 1: Slow heterolytic fission of C-Br bond to form planar (CH₃)₃C⁺ carbocation and Br⁻ (RDS) (2). Step 2: Fast attack of OH⁻ nucleophile on carbocation to form alcohol (1).	[4]
(c) Stereochemical Outcomes	SN2 mechanism on (R)-2-bromobutane gives complete inversion of configuration (Walden inversion) yielding 100% (S)-butan-2-ol, maintaining optical activity (2). In an SN1 mechanism, the planar carbocation intermediate can be attacked from either face with equal probability (50% front, 50% back) (2). This yields an equimolar (50:50) racemic mixture of enantiomers, resulting in complete loss of optical activity (optically inactive) (1).	[5]
(d) Effect of Doubling [OH⁻]	(i) 1-bromobutane (SN2): Rate doubles because the reaction is first order with respect to OH⁻ (1.5). (ii) 2-bromo-2-methylpropane (SN1): Rate remains unchanged because the reaction is zero order with respect to OH⁻ (OH⁻ is involved only in the fast step after the RDS) (1.5).	[3]

QUESTION 34: Phenolphthalein Kinetics & Integrated Laws (18 Marks)

*(a) Level of Response Evaluation	Level 3 (5–6 marks): Comprehensive comparative evaluation addressing all 3 methods with clear links to experimental precision: (1) Tangent method: Direct determination of instantaneous rates, but drawing tangents by hand has high subjective error and tangent errors propagate when plotting rate vs conc. (2) Half-life method: Fast and non-mathematical; constancy of $t_{1/2}$ uniquely identifies 1st order, but requires following the reaction across at least 2 full half-lives, and background absorbance drift can distort V_{∞}. (3) Integrated plots: Most rigorous; linear regression of $\ln A$ vs t (1st order) vs $1/A$ vs t (2nd order) uses all data points simultaneously, minimizing random scatter and identifying subtle deviations. Level 2 (3–4 marks): Discusses 2–3 methods with valid points. Level 1 (1–2 marks): Superficial description of one or two methods.	[6]
(b) Half-Life & Pseudo-Rate Constant	(i) A falls from 1.200 to 0.600 in 120 s ($t_{1/2}(1) = 120$ s); falls from 0.600 to 0.300 in 120 s ($240 - 120 = 120$ s) (2). Constant half-life proves reaction is first order in phenolphthalein (1). (ii) Gradient of $\ln A$ vs $t = (-1.551 - 0.182) / (300 - 0) = -1.733 / 300 = -5.78 \times 10^{-3} \text{ s}^{-1}$ (2). $k' = -\text{gradient} = 5.78 \times 10^{-3} \text{ s}^{-1}$ (1).	[6]
(c) Order in OH⁻ and True k	Doubling $[\text{OH}^-]$ (0.100 → 0.200 → 0.400 → 0.800) causes k' to double consecutively (1.15 → 2.31 → 4.62 → 9.23×10^{-3}) (2). Therefore, order $n = 1$ with respect to OH⁻ (1). Overall rate equation: $\text{Rate} = k[\text{P}^{2-}][\text{OH}^-]$ (1). $k = k' / [\text{OH}^-] = (1.15 \times 10^{-3}) / 0.100 = 0.0115$ (or 1.15×10^{-2}) (1). Units: $\text{dm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ (1).	[6]

QUESTION 35: Ozone Decomposition & Catalytic Cycles (18 Marks)

(a) Negative Order Meaning	(i) Overall order = $2 + (-1) = 1$ (first order overall) (1). (ii) A negative order means that increasing the concentration of oxygen retards / slows down the rate of decomposition (1). Oxygen acts as an inhibitor because it shifts the pre-equilibrium step backwards (1).	[3]
(b) Derivation of Rate Law	Step 2 is slow RDS: $\text{Rate} = k_2 [\text{O}_3][\text{O}]$ (1). Step 1 fast equilibrium: $k_1 [\text{O}_3] = k_{-1} [\text{O}_2][\text{O}]$ (2). Solve for intermediate oxygen atom concentration: $[\text{O}] = (k_1 / k_{-1}) [\text{O}_3] / [\text{O}_2]$ (1). Substitute $[\text{O}]$ into rate equation: $\text{Rate} = k_2 ((k_1 / k_{-1}) [\text{O}_3] / [\text{O}_2]) [\text{O}_3]$ (1). $\text{Rate} = (k_2 k_1 / k_{-1}) [\text{O}_3]^2 [\text{O}_2]^{-1} = k [\text{O}_3]^2 [\text{O}_2]^{-1}$ with $k = (k_2 k_1 / k_{-1})$ (1).	[6]
(c) Numerical Rate Calculations	(i) $\text{Rate} = (5.00 \times 10^{11}) * (4.00 \times 10^{-8})^2 / (8.00 \times 10^{-6})$ (1). $= (5.00 \times 10^{11}) * (1.60 \times 10^{-15}) / (8.00 \times 10^{-6}) = 1.00 \times 10^{-7} \text{ mol dm}^{-3} \text{ s}^{-1}$ (1). (ii) Stoichiometric ratio: 2 moles O₃ consumed produces 3 moles O₂ (1). $d[\text{O}_2]/dt = (3/2) * \text{rate of O}_3 \text{ decomposition} = 1.5 * (1.00 \times 10^{-7}) = 1.50 \times 10^{-7} \text{ mol dm}^{-3} \text{ s}^{-1}$ (1).	[4]
(d) Catalytic Destruction by Cl Radical	(i) Alternate two-step mechanism replaces high activation energy O₃ + O collision with two lower activation energy radical steps (2). Chlorine radicals react rapidly with O₃ to form ClO, which reacts rapidly with O atoms to regenerate Cl (1). (ii) Cl radical is consumed in Step 1 and regenerated in Step 2; it acts as a homogeneous catalyst and cancels out of the overall reaction (2).	[5]

QUESTION 36: Stopped-Flow Kinetics of Fe³⁺ and SCN⁻ (15 Marks)

(a) Stopped-Flow Technique	Pneumatic drive syringes rapidly force reactant solutions through a high-efficiency mixing chamber into an optical flow cell (2). Flow is stopped abruptly by a stopping syringe pushing against a mechanical limit switch, triggering spectrophotometric recording (1). Allows dead time to be reduced to < 2 milliseconds, capturing rapid initial kinetics before equilibrium is established (1).	[4]
(b) Determination of k_f, k_r and K_c	(i) $k_{\text{obs}} = k_f [\text{Fe}^{3+}] + k_r$ is a straight line $y = mx + c$ with slope = k_f and intercept = k_r (1). Slope $k_f = (29.60 - 6.10) / (0.0600 - 0.0100) = 23.50 / 0.0500 = 470 \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ (2). Units: $\text{dm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ (1). (ii) y-intercept $k_r = 6.10 - (470 \times 0.0100) = 6.10 - 4.70 = 1.40 \text{ s}^{-1}$ (2). Units: s^{-1} (1). (iii) $K_c = k_f / k_r = 470 / 1.40 = 336$ (or 335.7) $\text{dm}^3 \text{ mol}^{-1}$ (1).	[8]
(c) Colorimetry Filter Selection	Complementary filter: Blue filter (wavelength 440–470 nm) (1). The $[\text{Fe}(\text{SCN})]^{2+}$ complex is blood-red because it absorbs complementary blue light strongly (1). A blue filter ensures maximum absorbance change per mole of complex formed, maximizing signal sensitivity (1).	[3]

QUESTION 37: Kinetics of Alcohol Oxidation by $\text{Cr}_2\text{O}_7^{2-}$ (15 Marks)

(a) Colorimetric Monitoring	Orange dichromate(VI) ions ($\text{Cr}_2\text{O}_7^{2-}$) are reduced to green chromium(III) ions (Cr^{3+}) (1). Absorbance can be monitored at 440 nm (decrease in orange $\text{Cr}_2\text{O}_7^{2-}$) or at 630 nm (increase in green Cr^{3+}) (1). Continuous monitoring records absorbance without withdrawing or quenching aliquots (1).	[3]
(b) Order Deductions, Rate Law & k	Compare Exp 1 & 2: $[\text{Cr}_2\text{O}_7^{2-}]$ and $[\text{H}^+]$ constant; [propan-2-ol] doubles, rate doubles $\Rightarrow$ Order in propan-2-ol = 1 (1.5). Compare Exp 1 & 3: [propan-2-ol] and $[\text{H}^+]$ constant; $[\text{Cr}_2\text{O}_7^{2-}]$ doubles, rate doubles $\Rightarrow$ Order in $\text{Cr}_2\text{O}_7^{2-}$ = 1 (1.5). Compare Exp 1 & 4: [propan-2-ol] and $[\text{Cr}_2\text{O}_7^{2-}]$ constant; $[\text{H}^+]$ doubles (0.500 $\rightarrow$ 1.000), rate quadruples (1.60 $\rightarrow$ 6.40) $\Rightarrow$ Order in H^+ = 2 (2). Rate equation: $\text{Rate} = k[\text{CH}_3\text{CH}(\text{OH})\text{CH}_3][\text{Cr}_2\text{O}_7^{2-}][\text{H}^+]^2$ (1). $k = \text{Rate} / ([\text{propan-2-ol}][\text{Cr}_2\text{O}_7^{2-}][\text{H}^+]^2) = 1.60 \times 10^{-5} / (0.200 \times 0.0100 \times 0.500^2)$ (1). $= 1.60 \times 10^{-5} / (5.00 \times 10^{-4}) = 0.0320$ (or 3.20×10^{-2}) (1). Units: $(\text{mol dm}^{-3} \text{ s}^{-1}) / (\text{mol dm}^{-3})^4 = \text{dm}^9 \text{ mol}^{-3} \text{ s}^{-1}$ (1).	[8]
(c) Kinetic Isotope Effect Interpretation	The large rate decrease (6.7-fold slower) when C–H is replaced by C–D indicates a primary kinetic isotope effect (2). Deuterium is heavier than protium, so the C–D bond has a lower zero-point energy and higher bond dissociation energy (1). This proves conclusively that the C–H bond of the secondary alcohol carbon is cleaved in the slow rate-determining step of the mechanism (1).	[4]